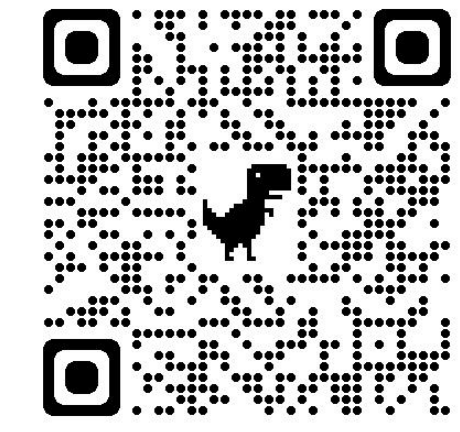




DPS-Net: Deep Polarimetric Stereo Depth Estimation

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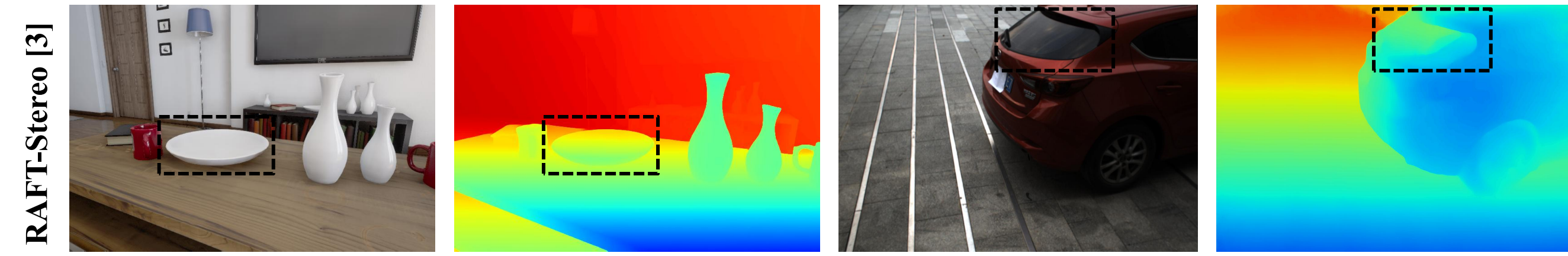
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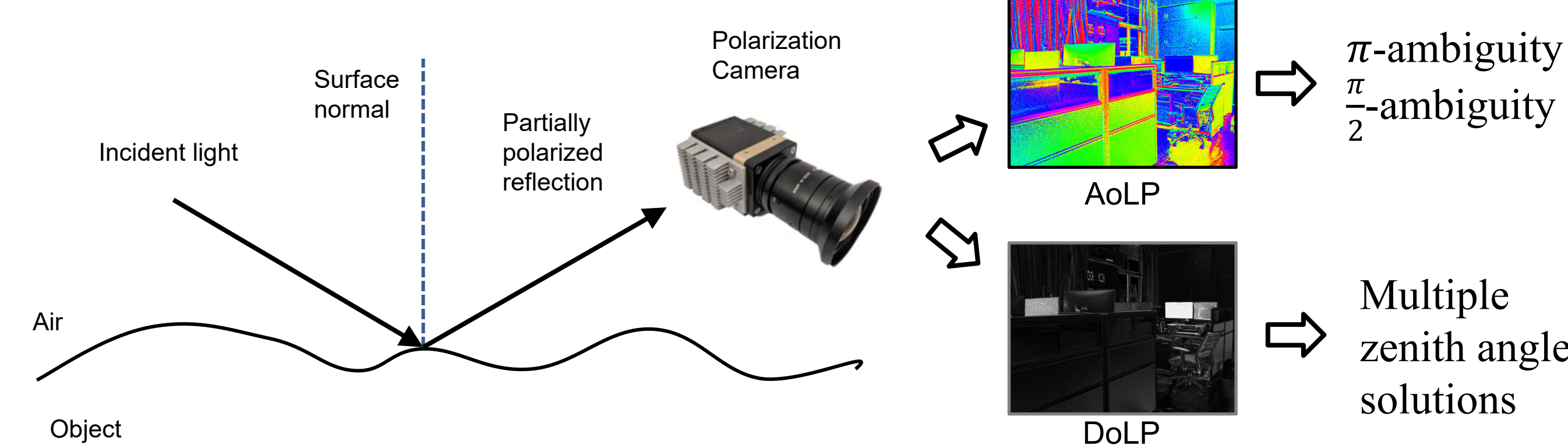
ICCV23
PARIS

Motivation

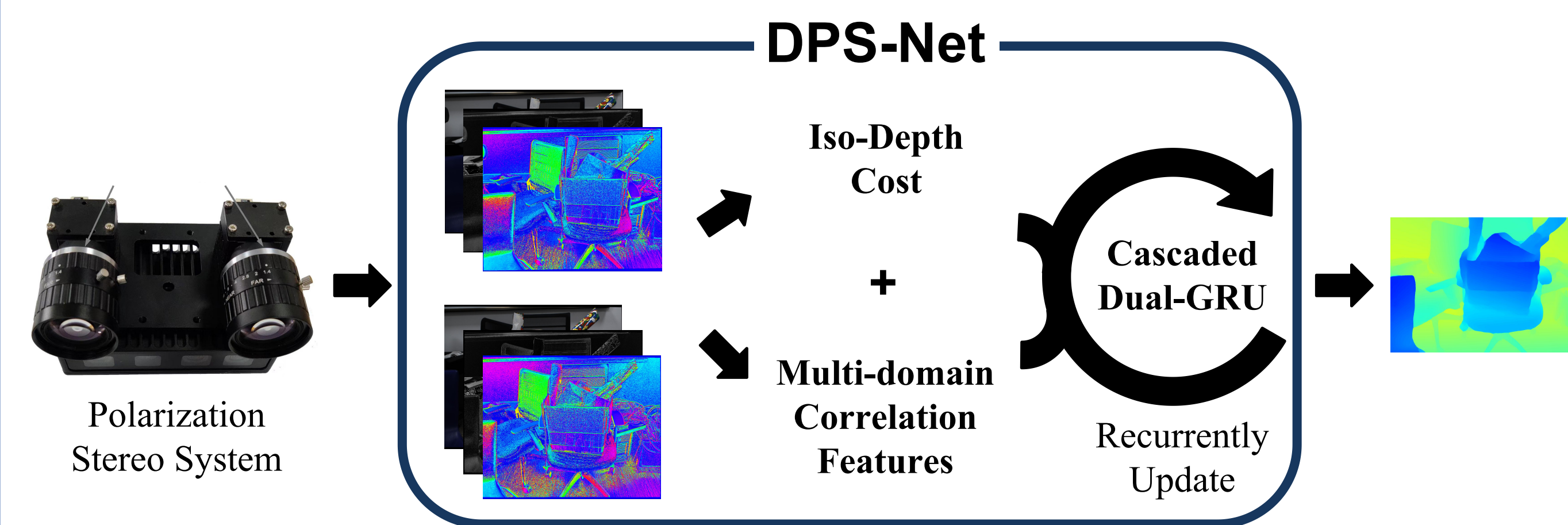
- RGB-based stereo matching struggles with **textureless areas**.



- Shape from polarization struggles with **ambiguities**.



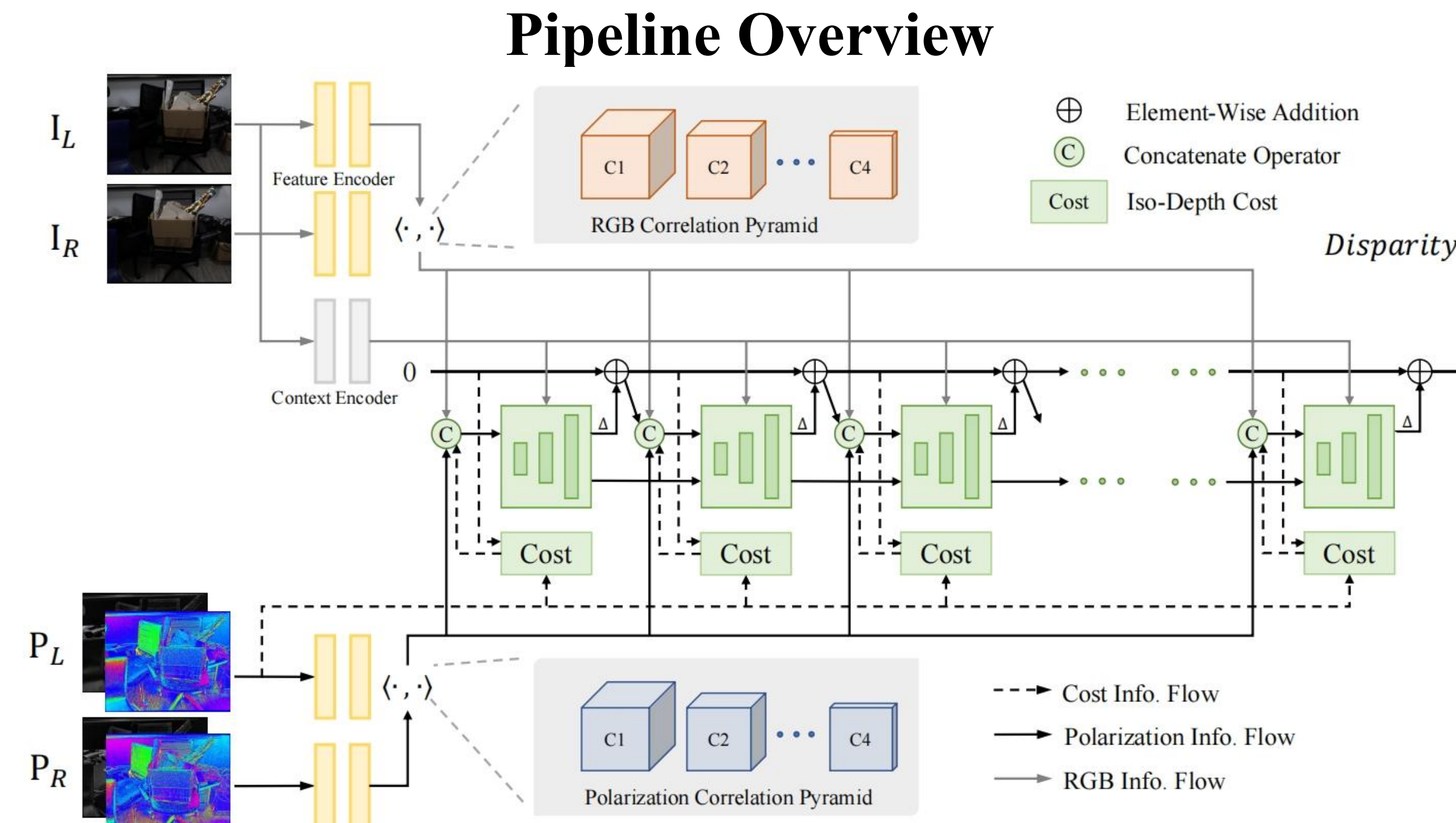
Our Solution



Reference

- [1] Liu et al. Local similarity pattern and cost self-reassembling for deep stereo matching networks. AAAI 2022.
- [2] Cheng et al. Hierarchical neural architecture search for deep stereo matching. NeurIPS 2020.
- [3] Lipson et al. Raft-stereo: Multilevel recurrent field transforms for stereo matching. 3DV 2021.
- [4] Zhu et al. Depth from a polarisation+ rgb stereo pair. CVPR 2019.
- [5] Lei et al. Shape from polarization for complex scenes in the wild. CVPR 2022.
- [6] Ba et al. Deep shape from polarization. ECCV 2020.

Method



- Extract the multi-domain features.
- Construct the multi-domain correlation pyramids.
- Integrate the iso-depth cost.
- Update the disparity recurrently by the GRU.

Iso-Depth Cost

We construct the unified iso-depth cost.

- Bypass both the zenith angle ambiguity and π -ambiguity of azimuth angle.
- Handle the $\pi/2$ -ambiguity of azimuth angle.

$$C_s(\varphi) = [\sin(\phi) \sin(\varphi) + \cos(\phi) \cos(\varphi)]^2,$$

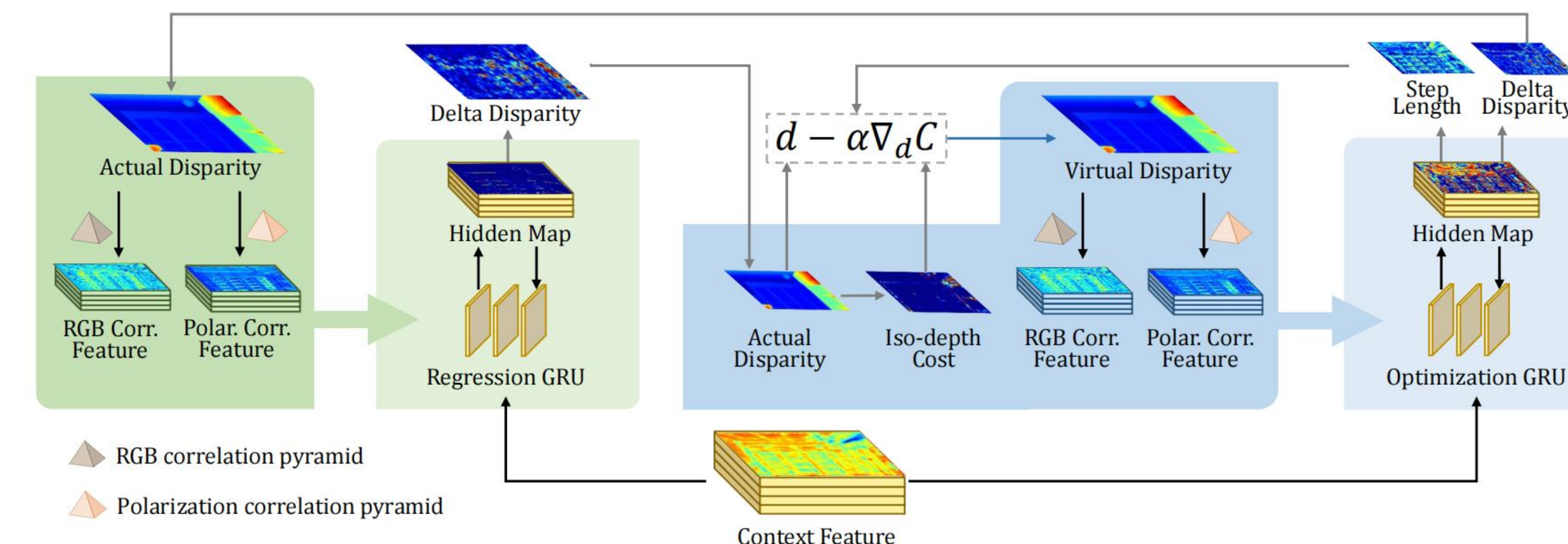
$$C_d(\varphi) = [\sin(\phi) \cos(\varphi) - \cos(\phi) \sin(\varphi)]^2,$$

$$C(\varphi) = \min\{C_s(\varphi), C_d(\varphi)\},$$

$$R(\varphi) = \arg \min\{C_s(\varphi), C_d(\varphi)\}.$$

$$\psi = \begin{cases} \varphi & \text{if polarized diffuse reflection dominates} \\ \varphi - \frac{\pi}{2} & \text{otherwise} \end{cases}$$

Cascaded Dual-GRU Architecture



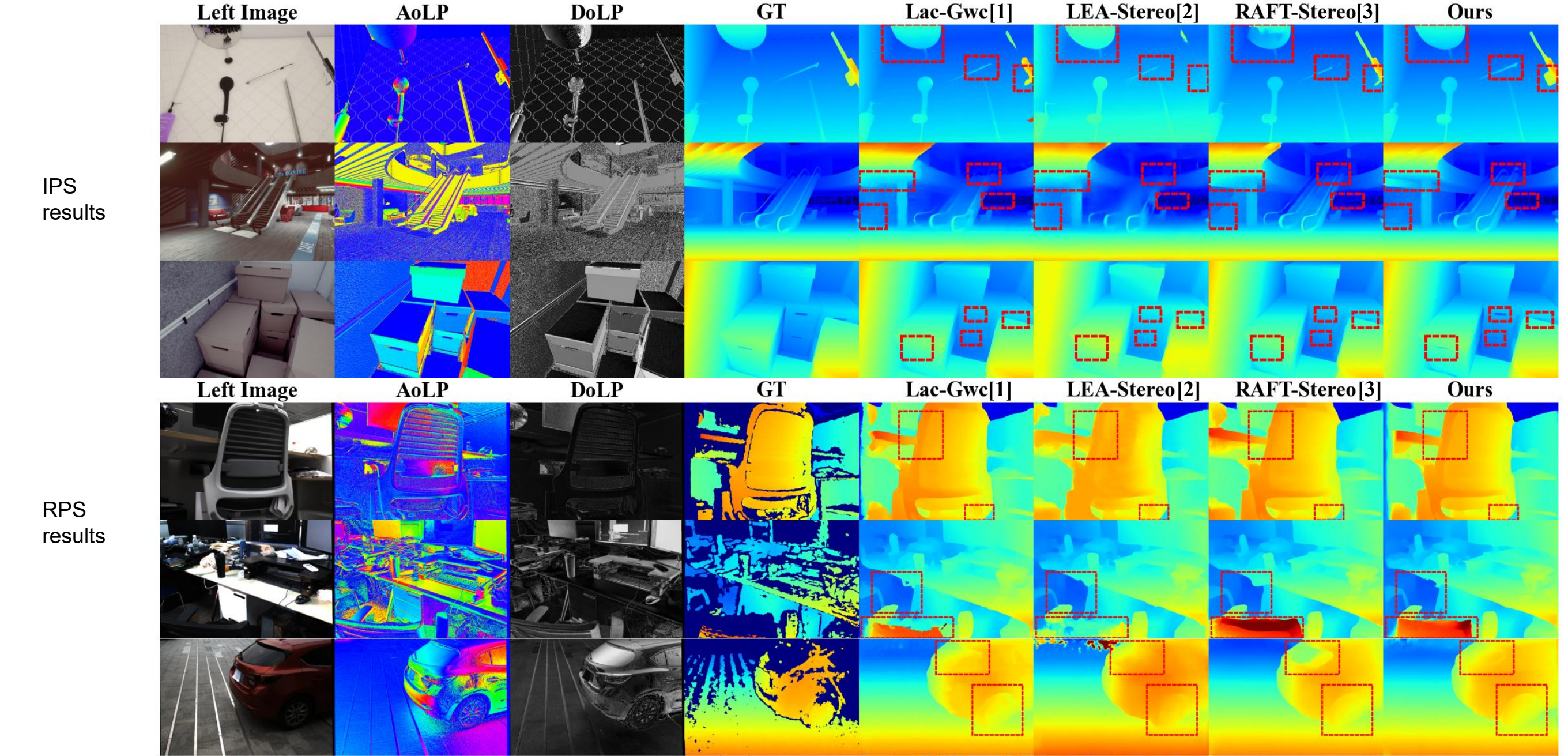
- In the regression GRU, the increment is regressed directly from the multi-domain correlation feature.
- In the optimization GRU, the disparity is further optimized based on the iso-depth cost.

Experiments

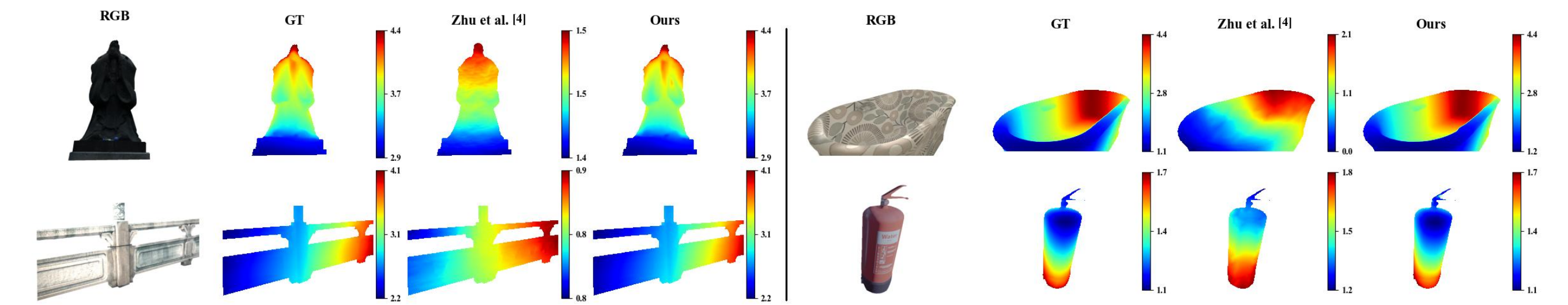
Quantitative Result

Comparison Results on the IPS dataset.				Comparison Results on the RPS dataset.		
Method	AvgErr	RMSE	bad 2.0	Method	AvgErr	bad 2.0 Runtime(s)
Lac-Gwc [1]	1.2135	3.4503	8.2061	Lac-Gwc [1]	0.6919	3.6735 0.702
LEA-Stereo [2]	1.6094	3.5475	11.247	LEA-Stereo [2]	0.7518	5.0487 0.359
RAFT-Stereo [3]	0.9266	2.6755	6.9791	RAFT-Stereo [3]	0.6807	3.8864 0.352
Ours	0.5790	1.8616	3.9705	Lac-Gwc-RBGP	0.6674	3.4693 0.798
				LEA-Stereo-RBGP	0.7517	5.1257 0.577
				RAFT-Stereo-RBGP	0.6244	3.4306 0.958
				Ours	0.6187	3.3541 0.255

Qualitative Depth Estimation Results



Qualitative Comparison with Traditional Polarimetric Method



Qualitative Normal Estimation Results

